



SOLITUDE 16MW SOLAR PV

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Project ID	2308
Monitoring period	01-January-2023 to 31-December-2023
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Version	5
VCS Standard Version	4.7
Prepared by	Voltas Yellow Ltd

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Solitude 16 MW solar PV (“the project”) consists of the construction and operation of a greenfield 16.328 MW_{DC} solar photovoltaic power plant in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses. It involves the setting up of photovoltaic (PV) panels which will capture solar energy and convey such energy to the convertor station in order to produce electricity exported to the national grid. Electricity in Mauritius is mainly generated from coal and heavy fuel oil, which is the baseline scenario and can also be considered as the scenario prior to the implementation of the project activity leading to considerable greenhouse gas (GHG) emissions. The project activity undertaken by project promoter Voltas Yellow Ltd will therefore substitute grid electricity by clean and renewable energy and cut down GHG emissions.

Voltas Yellow Ltd is 100% owned by Voltas Eco-Biotech Ltd. Voltas Eco-Biotech Ltd is 51% owned by GreenYellow (Mauritius) Ltd and 49% by Joonas & Co Ltd. Having this stated ownership sharing, GreenYellow (Mauritius) Ltd is the operating company for Solitude PV farm. In the following document, the company that will be mentioned is GreenYellow or GY, in the account of GreenYellow (Mauritius) LTD, as it is the company operating the project.

The proposed project consists of setting-up an amount of 59,376 solar PV modules with an installed capacity of 16.328 MW to produce electricity which is exported to the grid of the Central Electricity Board (CEB). The CEB in consequence will have to produce less electricity from coal or heavy fuel oil, which are non-renewable sources of energy. The energy production strategy implies a substantial reduction in the production of carbon and its release in the atmosphere, thereby reducing the associated greenhouse impact upon the atmosphere. The PV modules installed are of make CSUN275-60P, poly-crystalline silicon solar cells with good temperature coefficient enabling higher output in high temperature regions, which results in module efficiency of up to 16.94%. In the solar photovoltaic power plant, 24 modules are connected per string of 6.6 kW output power and connected to inverters & transformers’ shelter boxes (PTR1, PTR2 & PTR4 housing 3 inverters each, while PTR3 only features 2). The solar facility thus consists of 11 Inverter Units of make and model INGETEAM 1220TL, and 4 transformers of 440/22 kV of make and model SEA A0Bk.A 6.5km 22KV line connects the PV plant to the substation at Riche Terre (Jin Fei). The substation (22/66KV) is the delivery and metering point, including protections and control equipment, then conveying electricity to a 14MVA ABB transformer to rise power to 22/66KV. The monitoring equipment is composed of two class 0.2 meters for import/export measurement, one of which is used for back-up, located at CEB sub-station.

The construction stage of the project was completed in October 2018 and in its operation stage since then for a duration of 20 years. It is to be noted that the project Commercial Operation Date (COD) with the Central Electricity Board (CEB) was on the 19th of December 2018.

The project will generate approximately 28,882 tCO_{2e} emission reductions per year and 202,171 tCO_{2e} of emission reductions over the 7 years crediting period. During the fourth monitoring period 01-January-2023 to 31-December-2023, the project has operated satisfactorily generating 22.2 GWh of electrical energy and avoiding 21,847 tons of CO₂. However, the plant did not operate at its full potential during the month of May and October due to an earth fault detected in the 22 kV cable.

1.2 Audit History

The table below includes the audit history of the project.

Audit Type	Period	Program	VVB Name	Number of years
Validation and Verification	19-December-2018 – 30-April-2020	VCS	Earthood Services Private Limited	1 year and 4 months
Verification	01-May-2020 -- 31-December-2021	VCS	Carbon Check (India) Private Ltd.	1 year and 8 months
Verification	01-January-2022 – 31-December-2022	VCS	Earthood Services Private Limited	1 year
Verification	01-January-2023 – 31-December-2023	VCS	Earthood Services Private Limited	1 year
Total				5 years

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	Scope 1 – Energy Industries (renewable sources) – ACM0002: Large-scale
Project activity type	Renewable energy

1.4 Project Proponent

Organization name	Voltas Yellow Ltd
Contact person	Ramchand COONJUL
Title	Deputy Chief Operations Officer
Address	Beau Plan Business Park, North 1A Pamplemousses – Mauritius
Telephone	T: +230 260 5050 / M : +230 58 62 82 15

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

Email	rcoonjul@greenyellow.com
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1.5 Other Entities Involved in the Project

Organization name	GreenYellow SAS
Role in the project	Corporate support
Contact person	Sabrina MEDDAHI
Title	Global Impact Manager
Address	Tour Initiale, GreenYellow, 1, Terrasse Bellini 92800 La Défense - France
Telephone	+33 6 08 12 85 55
Email	smeddahi@greenyellow.com

1.6 Project Start Date

Project start date	19-December-2018
Justification	It is the date of commissioning, synchronization and start of GHG emission reductions

1.7 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	19-December-2018 to 18-December-2025

1.8 Project Location

Host Party: Mauritius

Region/State/Province: District of Pamplemousses

City/Town/Community: The site is located in Solitude, which is near the village Triolet.

The project geo-coordinates are Latitude: 20° 04'44"S; Longitude: 57° 32'30" E.



Figure 1: Localisation of project site

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources,	21.0

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.10.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.12 Sustainable Development Contributions

The project participants are confident that the proposed project activity will make significant impact on Mauritius sustainable development. A CSR program is developed annually to ensure local populations' needs and to consider all our stakeholders' points of view. In that framework, Solitude CSR program was voluntarily created and put in place since the beginning of 2022. Every year, actions with positive impacts for the Solitude community are organized and put in place. These actions are developed to respond to different needs in different areas like youth awareness, agriculture, and education.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	13.0	Tons of greenhouse gas emissions avoided or removed	Implemented activities to increase	During the monitoring period, the project has allowed to avoid the emission of 21,847 tCO ₂ .	The project generates electricity from a renewable and clean energy source, thus avoiding the dispatch of the same amount of electricity produced by fossil-fuelled power plants to Mauritius grid. Besides, the SO _x emissions from coal-fired power plants would also be reduced.

2)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	During the fourth monitoring period 01-January-2023 to 31-December-2023, the project has operated satisfactorily, generating 22.2 GWh of electrical energy.	The project promotes and stimulates the commercialization of grid-connected renewable energy technologies in Mauritius, in line with the Government's renewable energy policy. Adopted in 2011, the vision of Maurice Ile Durable (MID) targeted a 35% share of total energy needs to be fulfilled with renewable energies by 2025. In the Budget Speech 2021/22, the Government of Mauritius announced the Green Energy as a new economic pillar of Mauritius whereby the Renewable Energy (RE) target in the energy mix was increased from 40% to 60% by 2030 together with the announcement of the phasing out of coal in the generation of electricity by the same time.
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3)	7.b	7.b.1 Installed renewable energy-generating capacity in developing countries (in watts per capita)	Implemented activities to increase	No further changes this monitoring period	The project is estimated to bring clean and sustainable electricity to 10,000 Mauritian households. The project will improve energy self-sufficiency of the country which is currently heavily reliant on imported fossil fuels (above 85% according to the governmental Digest of Energy and Water statistics (Ministry of Finance and Economic Empowerment, 2017), alleviating the associated risks of price variations. The government has announced plans to increase use of renewable sources of energy for electricity generation to 60 percent by 2030, and to phase out the use of coal by 2030. It aims to do this through wind farms, solar energy, biomass, wave, and waste-to-energy projects.
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4)	8.3	Proportion of local/regional employment opportunities	Implemented activities to increase	No further changes this monitoring period.	The project contributes to the local employment throughout its building and operation phases, its workforce consisting of up to 175 workers during the construction phase and around 10 workers every year (this number can vary from one year to another) in the operations phase (<u>Refer to Appendix n° 7 “Staff Log”</u>). It will also induce indirect employment by increasing the competitiveness of local industry from reducing the country’s dependency on fossil fuels. Therefore, Mauritius Government is supportive of the project because the development of solar PV power is in accordance with the national criteria for sustainable development and national policies relating to energy resources and the environment, which will push forward the use of renewable and clean energy across the country.
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6)	4.7	Provide awareness to younger generation about climate change and energy transition	Implemented activities to increase	During the Monitoring Period, 183 children and students have been welcomed on Solitude Solar farm to visit it and understand how solar energy works.	Since 2022, children, students and local stakeholders are welcomed to the PV plant to understand the stakes of global warming and how solar energy could tackle it. Climate awareness is key to the energy transition and Solitude is a means to achieve it.
7)	1.4	1.4.1 Proportion of population living in households with access to basic services	Implemented activities to decrease	During the Monitoring Period, a donation of farming material has been done to Caritas Solitude – Lakaz Lespwar, to help them with their organic garden where 15 women are working daily.	A durable partnership with Caritas Solitude – Lakaz Lespwar, an organization that helps families in need, has been developed since 2020. Solidarity action campaigns among GreenYellow employees are conducted yearly to collect clothes, books and toys for children (200kg in 2022).

8)	2.4	2.4.1 Proportion of agricultural area under productive and sustainable agriculture	Implemented activities to increase	This activity is carried out every year. No further changes this monitoring period	Since the beginning of the project, the "Petits Planteurs" cooperative comes to plant and cultivate vegetables in a more sustainable way thanks to different methods that were developed by Vijaye Chuturdharee, Agricultural Engineer. These plantations cover one of the four zones of the PV plant. In addition to the vegetables collection, it also helps the maintenance of the plant in an ecological way. The benefits are mutual.
9)	4.4	Provide training in agro/bio-organic farming for women in agriculture	Implemented activities to increase	During the Monitoring Period, GreenYellow created and organized a training course in organic farming for 15 women from the Solitude community and farmers at Caritas Solitude. The objective was to guide the participants towards practices that are more respectful of the environment and their health. Refer to Appendix n° 8.	-

1.13 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the monitoring report.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

All the stakeholders were already identified at verification and no stakeholder make up has changed since validation.

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	The project proponent is in constant communication with the local stakeholders, mainly through Mr Vijaye Chuturdharee, Agricultural Engineer & Director of the planters' cooperative association. One of our project staff is on site every day. This enables us to collect requests and allow communication with the local population daily. In addition, as part of our Solitude CSR program, we have a long-term partnership with Caritas Solitude- Lakaz Lespwar, a local NGO working with local populations in difficulty.
Date(s) of stakeholder consultation	No stakeholder meeting has been conducted during the monitoring period. A stakeholder meeting will be held in 2024. The previous meeting was held on 06/07/2018 during the validation stage.
Communication of monitored results	During this monitoring period 01–January 2023 to 31–December 2023, no formal consultation meeting was held, and no other local stakeholder comments were received but regular contact has been maintained with Mr. Vijaye Chuturdharee over the year. Throughout the operation phase of the solar farm, two official environmental reports are prepared on a bi-annual basis which are also transferred to the stakeholders. This tool allows the stakeholders to follow the project's development and raise their queries, if they have any. A log sheet is also available on site whereby visitors and

	stakeholders are encouraged to register their remarks and grievances, if they have any. Regular exchanges take place with farmers belonging to the cooperative “Les Petits Planteurs”. Every year, they come and cut the grass that grows under the solar panels and take it away to feed their livestock.
Consultation records	Refer to Appendix n° 4 “ <u>Stakeholder Minutes of Meeting at Solitude 11072018 for Green Yellow Ltd (Final)</u> ”
Stakeholder input	No stakeholder input was received during the monitoring period.

2.1.3 Free, Prior, and Informed Consent

Consent	A first Local Stakeholder Consultation (LSC) meeting was held at the site office of Voltas Yellow Ltd on 06/07/2018, with the objective of gathering comments and concerns of stakeholders on the PV Based Power Generator by Voltas Yellow Ltd in Mauritius at Solitude, District of Pamplemousses, Mauritius. The stakeholders were invited by card and by a press advertisement in the local newspaper. Participants were invited to give their feedback in an open and transparent manner following a presentation on the proposed project activities. The public comments received during the LSC were answered during the 1-hour meeting and no negative comments were received. The consultation meeting began with an informative presentation delivered on Solar PV based power generation and its role in mitigating Global Warming. The phenomenon of Global Warming, the role played by Green House Gases (GHGs), the importance of the Kyoto Protocol and the Clean Development Mechanism (CDM) as mechanism for avoiding GHG emission were explained. Dr Revin Panray Beeharry (facilitator of the meeting) elaborated on the climate change process and elucidated means of combating such a threat by promoting a cleaner technology through the active consideration and the adoption of ‘Green Technologies’ such as solar PV. Dr Revin Panray Beeharry also stressed on the direct and indirect benefits accrued by such projects and allayed fears of any environmental pollution due to the project activities. Additionally, he explained that the project would add impetus to the socio-economic fabric of the Solitude and Triolet areas and help in sustainable development, as
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	promulgated by the Kyoto Protocol and United Nation Framework Convention on Climate Change. During the Question-and-Answer session, it was clearer to the audience as to how the PV project was less polluting and had no side effects compared to conventional ways of power generation. People also exchanged their views on the changes and benefits brought to the village by the project. The audience looked forward to such good development projects and pondered upon how such projects would fare in the future and whether this project would stand the test of time and last for at least 20 years. Refer to Appendix n° 4 for more details.
Outcome of FPIC	Stakeholders in general expressed their satisfaction at the initiative and welcomed the project. From their perspective, the project activity would accelerate socio-economic development in the area. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement. Kindly consult <u>“Stakeholder Minutes of Meeting at Solitude 11072018 for Green Yellow Ltd (Final)”</u> (Appendix n° 4) bearing no such remarks.

2.1.4 Grievance Redress Procedure

No grievances were raised during the monitoring period.

2.1.5 Public Comments

Until this monitoring period, no comments requiring further action were received.

However, all comments as part of the public stakeholder consultation are those mentioned above and enclosed in the appendix section.

Summary of comments received	Actions taken
N/A	N/A

2.2 Risks to Stakeholders and the Environment

Risk identified	Mitigation or preventative measure taken
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Risks to stakeholder participation	1) Herbicide treatment - Equipment circulation	- Herbicide treatment: treatment monitored closely with moderate amount used only on specific zones machines cannot access and where crops and stakeholders' cultivation are not affected. - Equipment circulation: Dedicated ways for machines to pass for panel cleaning or to access the different zones of the PV plant. No co-activity permitted by farmers when machines are operating to ensure safety of people.
Working conditions	No risk identified	Working conditions on our project is ensured by the Mauritian Law (Workers' Rights Act 2019) and GY Group Policies: 2) 02_ES-PO-002 HR&LaborPracticePolicy – Global 3) Appendix n°9 - EXTERNAL_CODE_OF_PROFESSIONAL_ETHICS_GreenYellow
Safety of women and girls	No risk identified	Safety of women and girls on our project is ensured by GY Group Policies: 4) 02_ES-PO-002 HR&LaborPracticePolicy – Global 5) Appendix n°9 - EXTERNAL_CODE_OF_PROFESSIONAL_ETHICS_GreenYellow 6) Appendix n° 2 - Politique Anti-harcèlement et non discrimination_202303 7) Appendix n° 10 - HSE GreenYellow Golden Rules

Safety of minority and marginalized groups, including children	No risk identified	No minority and marginalized groups, including children, are impacted by the project. This is moreover ensured by GY Group Policies: 8) 02_ES-PO-002 HR&LaborPracticePolicy - Global 9) Appendix n° 9 - EXTERNAL_CODE_OF_PROFESSIONAL_ETHICS_GreenYellow 10) Appendix n° 10 - HSE GreenYellow Golden Rules
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	Noise: No risk identified as no significant source of noise present during operations as there are no rotating machines at the PV Plant. Air quality: No risk identified because there are no dust, exhaust or process emissions due to the operation of the PV Plant Water and Irrigation System: No risk identified Waste: No risk identified as the project does not generate significant quantities of waste during its operations phase. There is little domestic solid waste generated on site during the operation phase of the project. The waste generated comes from packaging materials	

	such as plastics, cardboard, and broken parts. The waste is stored in a suitable container for recycling or disposal at the nearest Municipality's waste management facility. In addition to this, green waste is generated through grass cutting. The grass is left on the ground for composting.	
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2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	No discrimination or sexual harassment has occurred, and none will be permitted. GreenYellow commitment in terms of discrimination and sexual harassment is formalized through this policy: <u>"Politique Anti-harcèlement et non discrimination 202303"</u> (Appendix n° 2)
Management experience	No new entity is now involved in project design or implementation.
Gender equity in labor and work	GreenYellow commitment in terms of gender equity in labor and work is formalized through this charter: <u>"GreenYellow Ethics charter"</u> (Appendix n° 1)
Human trafficking, forced labor, and child labor	GreenYellow commitment in terms of gender equity in labor and work is formalized through this charter: <u>"GreenYellow Ethics charter"</u> (Appendix n° 1)

2.3.2 Human Rights

GreenYellow commitment in terms of Human Rights is formalized through this charter: "GreenYellow Ethics charter" (Appendix n° 1).which is in line with applicable international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples.

2.3.3 Indigenous Peoples and Cultural Heritage

Due to its location, the project is not concerned by indigenous people rights and cultural heritage.

2.3.4 Property Rights

Disputes over rights to territories and resources	N/A
Respect for property rights	Legal contracts have been signed between Voltas Yellow Ltd and each stakeholder before implementation of project, to protect and preserve the property rights of stakeholders, IPs, LCs, and customary rights holders. The contracts will be viewed at the audit of the project.

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	There is no benefit-sharing plan under the contracts, simply the right to occupy the land for Voltas Yellow in exchange for the payment of rent to the owners.
Benefit sharing during the monitoring period	N/A

2.4 Ecosystem Health

An Environmental Impact Assessment (EIA) was required for the solar power plant as a “Power Generating Plant”, and it was conducted in line with the ESPA requirements. An EIA license has been obtained on September 5th, 2017 from the Ministry of the Environment and Sustainable Development for the setting-up and operation of the solar PV farm.

We have submitted eight EMRs (bi-annual reports) for the operational phase of the Photovoltaic Power Generating Plant of Capacity 16.3 MW till date, in accordance with section 18(2) (I) of the Environment Protection Act 2002 and clause 4 of the EIA licence ENV/DOE/EIA/1716 granted on 5th September 2017 for the proposed Photovoltaic Plant at Solitude.

	Risk identified	Mitigation or preventative measure taken during the monitoring period
Impacts on biodiversity and ecosystems	No risk identified	

Soil degradation and soil erosion	No risk identified because vegetation cover is planted to capture the excess run-off and prevent erosion.	
Water consumption and stress	Use of water for panel cleaning	Clear water with no chemical is used to clean solar panels across the different zones of the solar farm. At the end of every maintenance period, the service provider measures the total quantity of water used per water tank in every zone of the solar farm and counts the number of water tanks deployed across the farm for the cleaning purpose, and shares the final water consumption value with GY. The quantity of water used is recorded every year in our yearly CSR report and its objective is to calculate GY impact on the environment and use of natural resources.
Usage of fertilizers	No risks identified as fertilizers are used in controlled amounts and in only a few areas of solar farm where farming is practiced enhancing crop growth.	

2.4.1 Rare, Threatened, and Endangered species

The project is not located in, or adjacent to habitats for rare, threatened, or endangered species.

2.4.2 Introduction of species

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A	N/A	N/A	N/A

Existing invasive species	Mitigation measures to prevent spread or continued existence of invasive species
N/A	N/A

2.4.3 Ecosystem conversion

This project is not an ARR, ALM, WRC or ACoGS project.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The commission date of the project activity is 19-December 2018, which is also the start date of GHG emission reductions. During the monitoring period 01-January 2023 to 31-December 2023, the project has operated continuously and satisfactorily, generating a total net energy delivered to the grid equal to 22.2 GWh. The only noteworthy event during the monitoring period involved the shutdown of the plant due to the earth fault on the 22 kV cable in May and October. No changes on the project activity or applicable legislation have occurred during this monitoring period.

3.2 Deviations

3.2.1 Methodology Deviations

No methodology deviations occurred during this monitoring period.

3.2.2 Project Description Deviations

No methodology deviations occurred during this monitoring period.

3.3 Grouped Projects

This is not a grouped project.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes

☒ No

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ e/MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year
Source of data	Representatives of Ministry of Energy of Mauritius
Value applied	0.9915
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{grid,OM,y}
Data unit	tCO ₂ e/MWh
Description	Operating Margin CO ₂ emission factor for grid connected power generation in year y calculated using the CDM "Methodological Tool to calculate the emission factor for an electricity system".
Source of data	IGES List of Grid Emission Factors
Value applied	1.0282
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{grid,BM,y}
Data unit	tCO ₂ e/MWh
Description	Build Margin CO ₂ emission factor for grid connected power generation in year y calculated using the CDM "Methodological Tool to calculate the emission factor for an electricity system".
Source of data	IGES List of Grid Emission Factors
Value applied	0.8814
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions

Comments	-
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Data / Parameter	The percentage share of total installed capacity of solar PV
Data unit	%
Description	The percentage share of total installed capacity of the solar PV in the total installed grid connected power generation capacity in the host country.
Source of data	http://publicutilities.govmu.org/ Statistics Mauritius (2018): Digest of Water and Energy Statistics - 2017, Port Louis.
Value applied	2.6
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Additionality demonstration
Comments	-

Data / Parameter	The total installed capacity of solar PV
Data unit	MW
Description	The total installed capacity of grid-connected solar PV in the host country.
Source of data	http://publicutilities.govmu.org/
Value applied	21.2
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Additionality demonstration
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y

Source of data	Measured directly with electricity meter(s) at Interconnection Facilities CEB JINFEI Substation																							
Description of measurement methods and procedures to be applied	-																							
Frequency of monitoring/recording	Continuous measurement and at least monthly recording																							
Value monitored	<table><tr><td>2023</td><td>TOTAL</td></tr><tr><td>21,847</td><td>21,847</td></tr></table>	2023	TOTAL	21,847	21,847																			
2023	TOTAL																							
21,847	21,847																							
Monitoring equipment	Electricity outputs are electronically recorded, stored and invoiced monthly to CEB by Voltas Yellow Ltd, based on main meter below (or back-up in case of main meter unavailability or default). The measurement devices of net electrical energy delivered to CEB grid are both located 20/63kV at JINFEI substation and feature the following specifications: <table><tr><td>Meter designation:</td><td>Main</td><td>Back-up</td></tr><tr><td>Meter ownership:</td><td>CEB</td><td>CEB</td></tr><tr><td>Make:</td><td>EDMI</td><td>EDMI</td></tr><tr><td>Model:</td><td>MK6E</td><td>MK6E</td></tr><tr><td>Accuracy class:</td><td>0.2S (±0.02%)</td><td>0.2S (±0.02%)</td></tr><tr><td>Serial Number:</td><td>16538506</td><td>16538507</td></tr><tr><td>Calibration date:</td><td>22.06.2020</td><td>22.06.2020</td></tr></table>			Meter designation:	Main	Back-up	Meter ownership:	CEB	CEB	Make:	EDMI	EDMI	Model:	MK6E	MK6E	Accuracy class:	0.2S (±0.02%)	0.2S (±0.02%)	Serial Number:	16538506	16538507	Calibration date:	22.06.2020	22.06.2020
Meter designation:	Main	Back-up																						
Meter ownership:	CEB	CEB																						
Make:	EDMI	EDMI																						
Model:	MK6E	MK6E																						
Accuracy class:	0.2S (±0.02%)	0.2S (±0.02%)																						
Serial Number:	16538506	16538507																						
Calibration date:	22.06.2020	22.06.2020																						
QA/QC procedures to be applied	Cross check of measurement results with records for sold electricity. Meter Laboratory of CEB is solely responsible for the selection, installation, calibration, servicing, testing, and repairing of all CEB energy meters (CEB Main Meter & CEB Back-up Meter). Voltas Yellow Ltd may install at its own expense a backup electric metering device (Seller Back-up Meter). As per PPA §12.1.4, CEB shall inspect each CEB Meter upon installation and at least once every year thereafter. CEB shall check the certification of CEB Meters through an accuracy test at least once every 4 (four) years thereafter or at any time the readings of Net Energy from the CEB Meter and Seller Back-up Meter differ by an amount greater than 0.5%.																							
Purpose of the data	Calculation of baseline emissions																							
Calculation method	-																							

Comments

The energy meters at the Jin Fei substation meters have been inspected and load tests have been carried out on the meters to confirm their correctness, as per schedule 12.1.4 of the Energy Supply and Purchase Agreement on 14th February 2023. See “[Meter Inspection](#)” (Appendix n° 3)

4.3 Monitoring Plan

The proposed project activity monitoring plan complies with the methodology ACM0002 Grid-connected electricity generation from renewable sources, Version 21.0 whereby it is stated that: As per ACM0002 provisions for record handling, all data collected as part of monitoring is archived electronically and kept at least for 2 years after the end of the last crediting period. All measurements are conducted with calibrated measurement equipment according to relevant industry.

Indeed, the quantity of net electricity generation supplied by the project plant to the grid is reliably monitored through calibrated electricity meters and cross-checked with sales records as part of quality assurance/quality control measures on top of CEB diligence. Besides, the reading of the main meter is compared with the check meter at the point of injection.

Data is also daily monitored, analyzed and cross-checked by the monitoring team through Quantum (QOS Energy) which is a web-based energy management platform that is in real-time communication with the equipment on site. The data acquisition equipment and installation are subject to regular maintenance and verifications as per the equipment manuals and the meters, to testing as per the ESPA, as stipulated in section 4.

Monitoring organization:

The Deputy Operations Director of GY coordinates and endorses the overall responsibility for all CDM monitoring of the project, including:

- Develop, approve, execute, and improve the CDM Monitoring/Reporting Procedures;
- Organize in-house seminar to inform and train the company staff to the monitoring procedures;
- Ensure that instrumentations and devices are available and properly suited to efficiently perform the monitoring;
- Communicate and coordinate the monitoring work of all business units;
- Validate and electronically archive all monitoring data on a monthly basis throughout the crediting period (and conserve it at least for 2 further years);
- Calculate and report the emission reductions; and
- Coordinate the DOE work during the verification audit.

The Director might appoint a CDM coordinator to delegate to him the above specific tasks of monitoring supervision. The Technical/Engineering/Maintenance Department consisting of plant technicians will undertake the technical actions required by the monitoring plan, under the Directors' authority, to collect and record related data.

The Accounting/Sales Department (Chief Financial Officer) will crosscheck, reconcile, or consolidate data with multiple sources whenever possible. At minimum, data obtained from the electricity meters is to be crosschecked with the electricity sales receipts. This kind of reconciliation activity will be recorded properly as DOE may request for such information during the verification.

Data collection, consolidation and results analysis is undertaken by a dedicated team adequately trained, aware of CDM requirements. O&M workflow manual enforced on-site further includes all updated procedures in case of mistake/breakdowns, emergency disconnections and cyclone warnings.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The monitoring period for which GHG emission reductions were achieved spans from 01/01/2023 to 31/12/2023. Baseline emissions are calculated according to the methodology as the product of (i) the quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y ($EG_{\text{facility},y}$ = Exports – Imports, in MWh/yr) and (ii) the combined margin CO₂ emission factor for grid connected power generation in year y ($EF_{\text{grid},\text{CM},y}$):

$$BE_y = EG_{\text{PJ},y} \times EF_{\text{grid},\text{CM},y}$$

Where:

BE_y	Baseline emissions in year y (t CO ₂ /yr)
$EG_{\text{PJ},y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EF_{\text{grid},\text{CM},y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO ₂ /MWh)

$EG_{\text{PJ},y} = EG_{\text{facility}}$ i.e. the quantity of net electricity generation supplied by the project plant to the grid, as monitored and displayed below on a monthly basis.

$EF_{\text{grid},\text{CM},y} = 0.9915$ tCO₂/MWh, as the Project Proponent have calculated, the DOE validated and the CDM-EB registered.

Year	Imports	Exports	$EG_{\text{facility},y}$	$EF_{\text{grid},\text{CM},y}$	BE_y
	kWh	kWh	MWh	tCO ₂ /MWh	tCO ₂
Jan-2023	8,824	1,903,511	1,895	0.9915	1,878
Feb-2023	14,663	1,928,162	1,914	0.9915	1,897
Mar-2023	9,321	2,121,689	2,112	0.9915	2,094
Apr-2023	9,332	1,786,303	1,777	0.9915	1,761
May-2023	8,428	1,035,221	1,027	0.9915	1,018
June-2023	10,032	1,667,634	1,657	0.9915	1,642
July-2023	10,954	1,724,526	1,714	0.9915	1,699
Aug-2023	10,248	2,054,496	2,044	0.9915	2,026

Sept-2023	9,370	2,191,778	2,182	0.9915	2,163
Oct-2023	8,769	1,639,087	1,631	0.9915	1,617
Nov-2023	9,061	2,071,805	2,062	0.9915	2,044
Dec-2023	8,954	2,034,912	2,026	0.9915	2,008
TOTAL	117,956	22,159,124	22,041		21,847

5.2 Project Emissions

No project emissions are expected as the project activity only involves renewable electricity generation from the solar power plant: $PE_y = 0$.

5.3 Leakage Emissions

As stated in the applicable methodology, no leakage emissions are considered.

5.4 GHG Emission Reductions and Carbon Dioxide Removals

The monitoring period for which GHG emission reductions were achieved spans from 01/01/2023 to 31/12/2023, as it follows:

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
01-January-2023 to 31-December-2023	21,847	0	0	21,847		21,847
Total	21,847	0	0	21,847		21,847

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
01-May-2020 to 31-Dec-2021	48,765	39,578	19%	Lower plant performance due to power phase damage (manufacturing defect)
01-Jan-2022 to 31-Dec-2022	28,978	23,557	19%	Lower plant performance due to power phase damage (manufacturing defect)
01-Jan-2023 to 31-Dec-2023	28,777	21,847	24%	Lower electricity generation in 2023 due to earth fault on 22 kV cable, in May and October.

Total	106,520	84,982	20%	
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